fourth—we can just think of the past and the future as distant points on the same fabric, stretching across the cosmos from its infancy to its end. To someone sitting at a different point on this fabric, an event that is part of the future to us might be the distant past to them. And the light (or any information) from an event that we won't see for millennia is already streaming across spacetime toward us "now." Is that event in the future, or the past, or, perhaps, both? It all depends on perspective.

As mind-bending as it is to contemplate if you're used to thinking in a 3D world,[fn3] to astronomers, the noninfinite speed of light is a fantastically useful tool. It means that instead of looking for mere clues to the distant past of the cosmos—its traces and remnants—we can just look at it directly and watch it change over time. We can peer at the universe at an age of just three billion, during the renaissance of star formation, when galaxies were bursting with light (if not art and philosophy), and we can see how that shine has dimmed in the intervening eons. We can look even farther back, and see matter swirling into supermassive black holes in a universe less than 500 million years old, when starlight had only just begun to penetrate the darkness between galaxies.

Soon, with new space telescopes, we will be able to observe some of the first galaxies to form in the cosmos—those that formed when the universe was only a few hundred million years old. But if those galaxies were the first, what happens if we look farther back than that? Can we look so far away that there are no galaxies yet? We have plans to do so. Radio telescopes being built now may be able to see the material the first galaxies were born from, by exploiting a fortuitous interaction between light and hydrogen. By looking directly at the hydrogen, the matter that will one day become stars and galaxies, we can watch the very first structures in the universe form.